

AGV Route

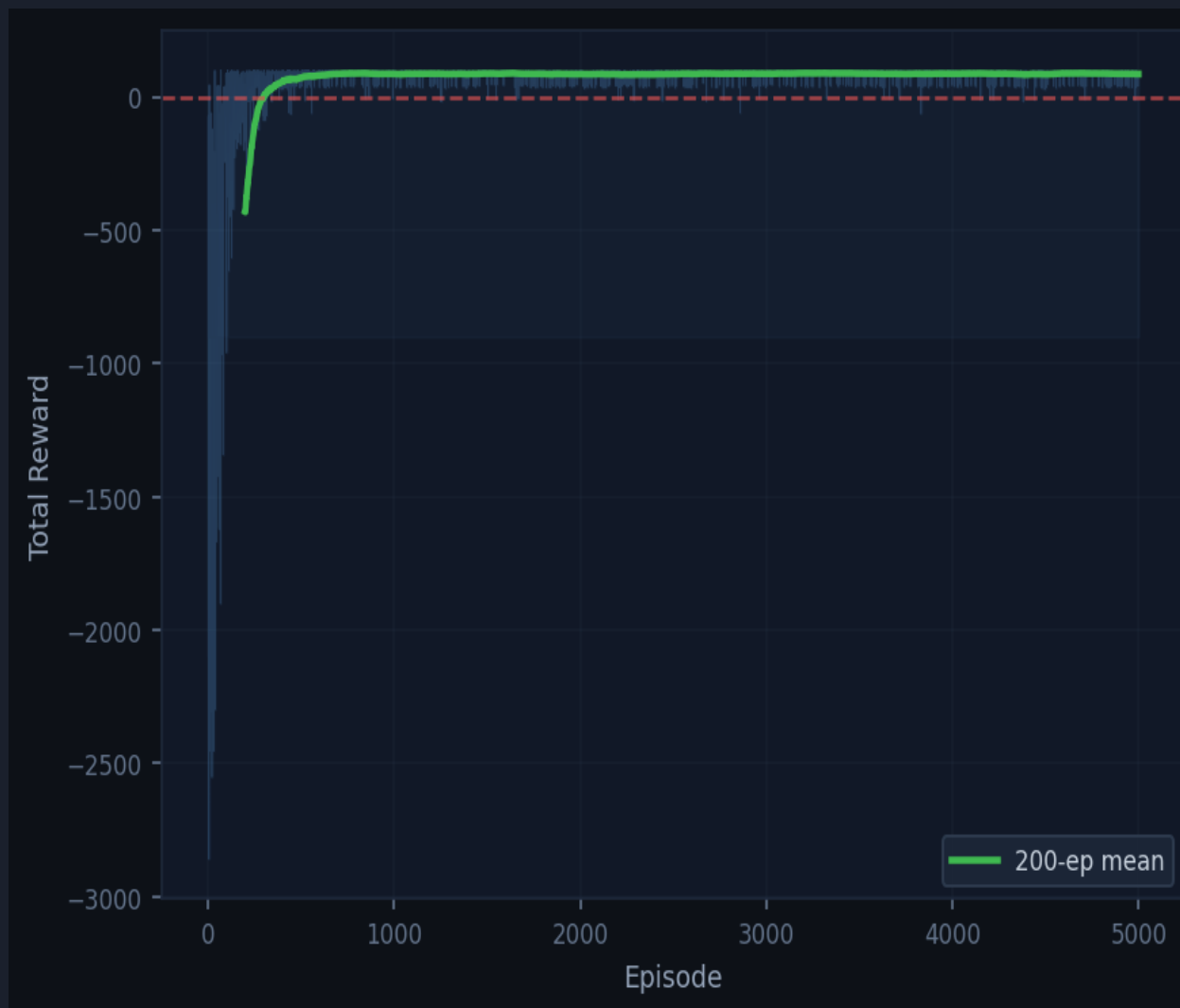
RL model

*Tabular Q-Learning · Factory Floor
Autonomous Navigation from Scratch*

Q-LEARNING

REINFORCEMENT RL

NUMPY + MATPLOTLIB



5,000

Training Episodes

-820

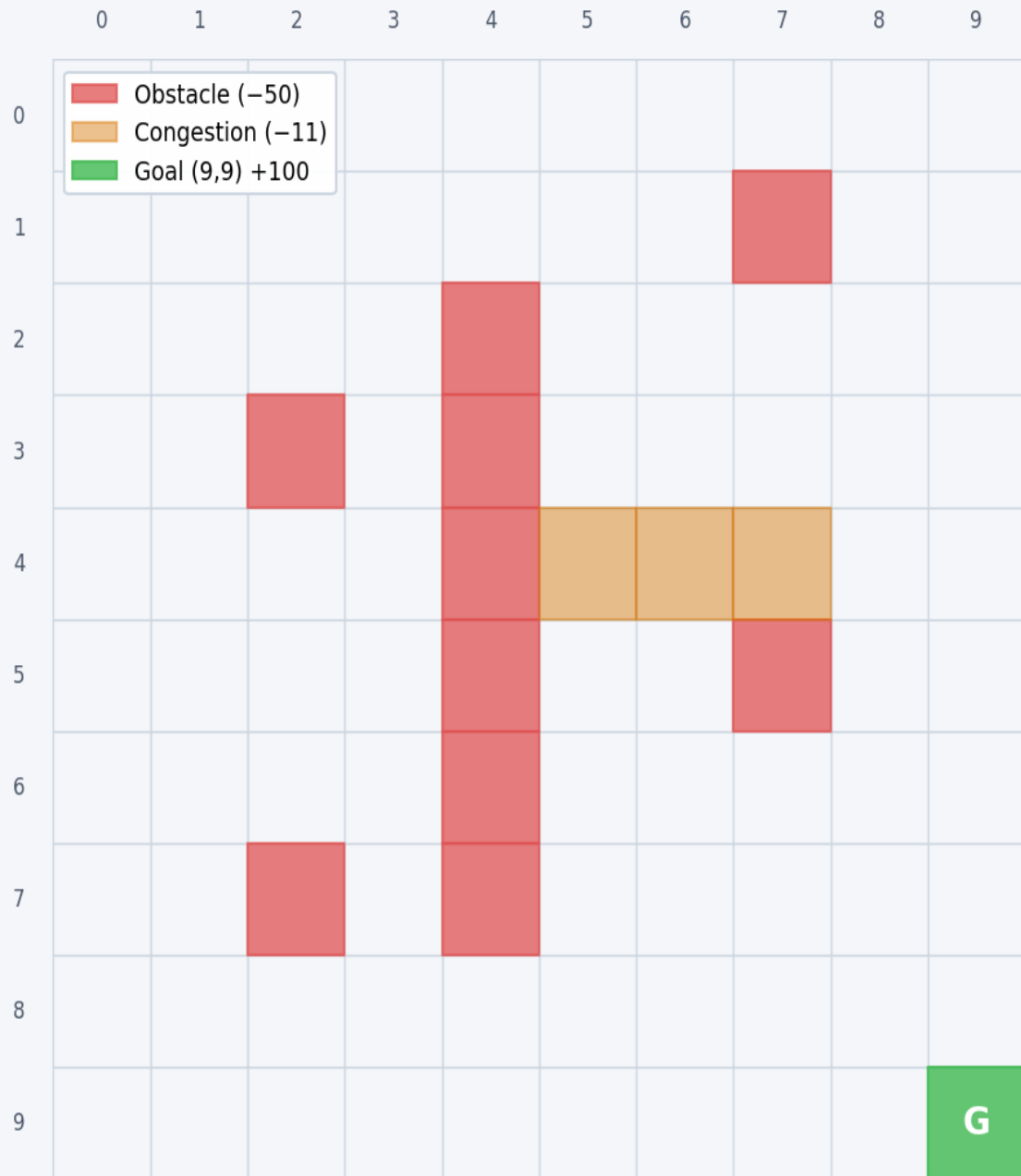
Ep 1 Mean Reward

85.3

Final Mean Reward

+906

Total Improvement



The negative step cost is not incidental — it is the pressure that produces the shortest path, not just any path.

Episode 0 — The World Before Training

The agent receives a world, a reward signal, and nothing else.

Grid	10 × 10 = 100 states
Actions	Up · Down · Left · Right
Obstacles	10 fixed (walls, equipment)
Congestion	3 high-traffic aisles
Goal	Delivery dock at (9, 9)
Start	Random valid cell each episode

Reward Function

Reach goal (9,9)

+100

Hit obstacle / wall

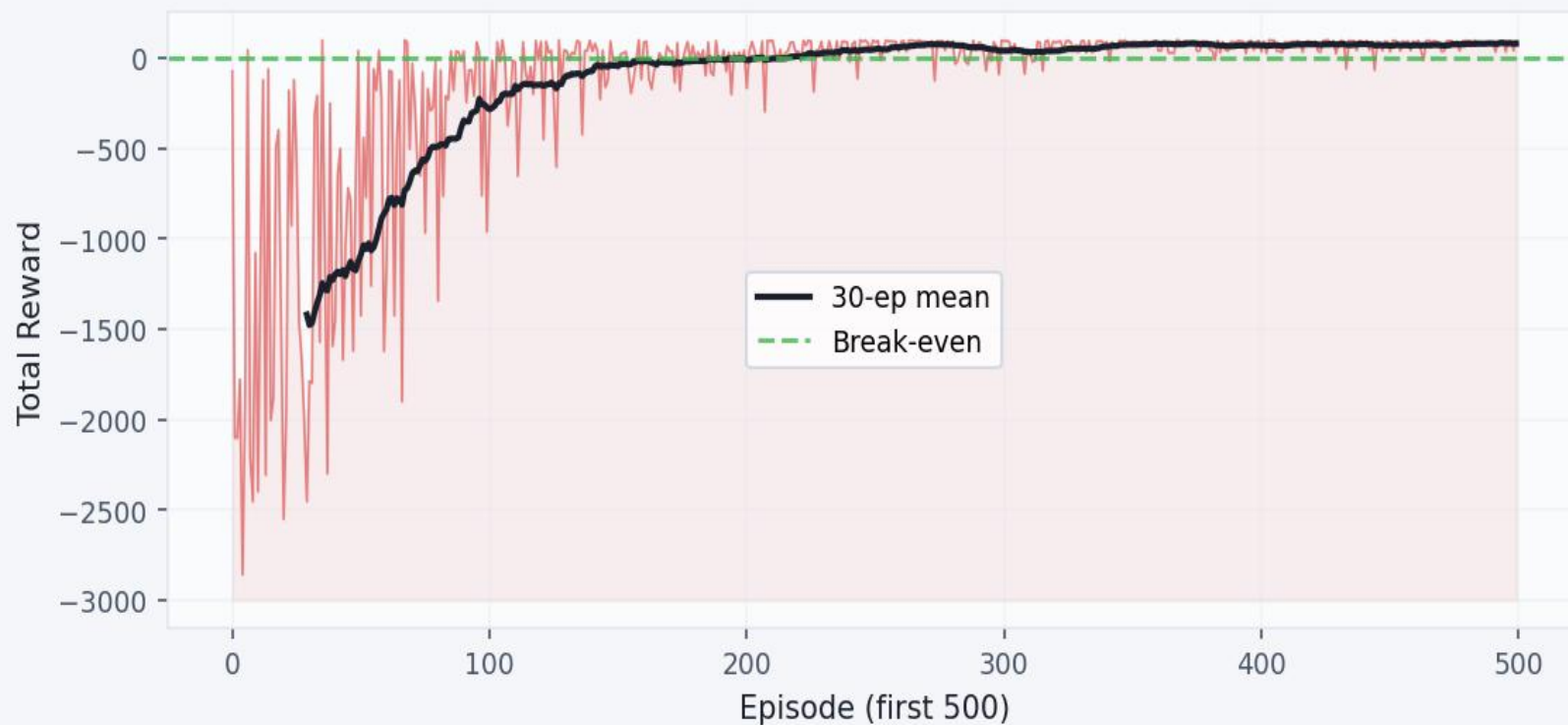
-50
Learning curve

Congestion crossing

Normal move

Episodes 1–100 — The Agent Knows Nothing

Q-table initialized at zero. Every decision is random. Every wall teaches.



-820.8

Mean Reward
Ep 1–100

$\epsilon = 1.0$

Exploration rate
(fully random)

-50×

Wall penalty
per collision

0

Q-table
initial values

The update that fires after every single step:

$$Q(s,a) \leftarrow Q(s,a) + \alpha \cdot [r + \gamma \cdot \max_{a'} Q(s') - Q(s,a)]$$

$\alpha = 0.10$ (learning rate) $\gamma = 0.95$ (discount factor) TD error = what we expected – what happened

Episode 1: the agent is not failing. It is collecting the evidence it needs to stop failing.

Episode 299 — The Turning Point

The 200-episode rolling mean crosses zero. The agent breaks even — and never looks back.

Early episodes — Explore

$\epsilon = 1.00 \rightarrow$ random action every step

The agent wanders — but builds a map of consequences.
Every wall collision writes -50 into Q(s,a).
Every goal arrival writes +100.

Late episodes — Exploit

$\epsilon = 0.05 \rightarrow$ best Q-value 95% of the time

The agent trusts its learned table.
Random exploration drops to 5% — kept only to detect environment changes.

ϵ decay: 1.00×0.995 per episode $\rightarrow \epsilon = 0.05$ floor at episode ~1,000

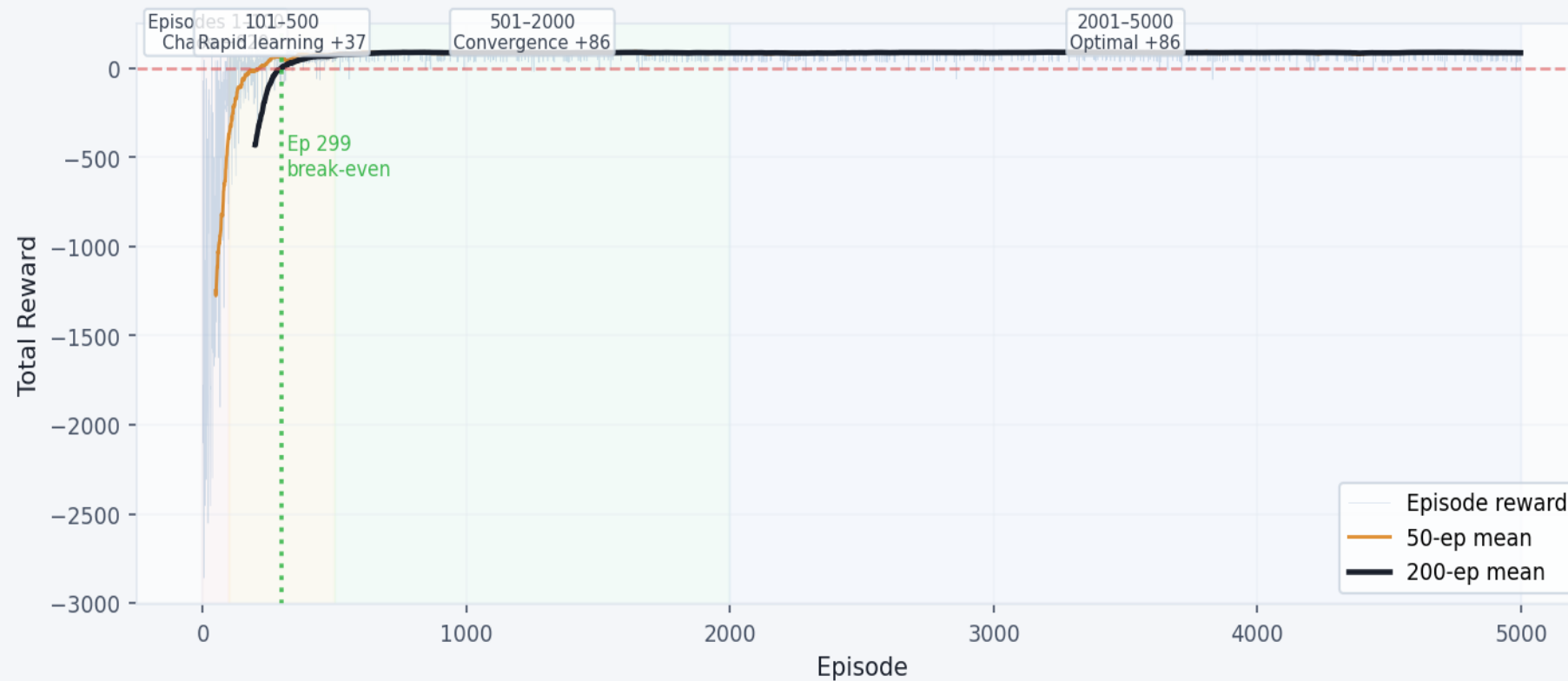
Hyperparameter Configuration

alpha (α)	0.10	Learning rate — how fast Q-values update per step
gamma (γ)	0.95	Discount — rewards 18 steps away still worth 40%
epsilon start	1.00	Pure exploration — agent starts completely blind
epsilon min	0.05	Minimum exploration — 1 in 20 moves stays random
epsilon decay	0.995	Multiplicative decay — reaches floor at episode ~1,000
episodes	5,000	Training budget — sufficient for 100-state environment

Episode 299: the 200-episode rolling mean crosses zero. Every episode after this is net-positive.

Episodes 500–5,000 — The Policy Takes Shape

From -820 to +85 across 5,000 episodes of trial, consequence, and adaptation.



Total swing:

-820.8



+85.3

over 5,000 episodes · +906 reward points · break-even at episode 299

Ep 1–100

ϵ : 1.00 $\rightarrow$ 0.61

-820.8

Ep 101–500

ϵ : 0.61 $\rightarrow$ 0.08

+37.0

Ep 501–2K

ϵ : 0.08 $\rightarrow$ 0.05

+86.2

Ep 2K–5K

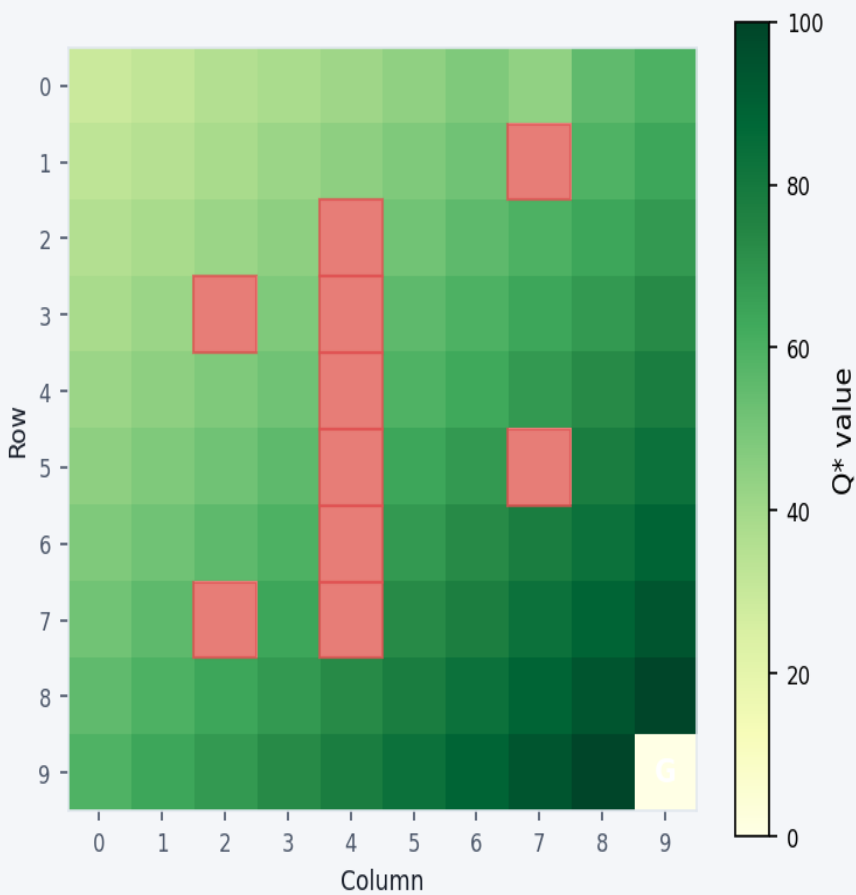
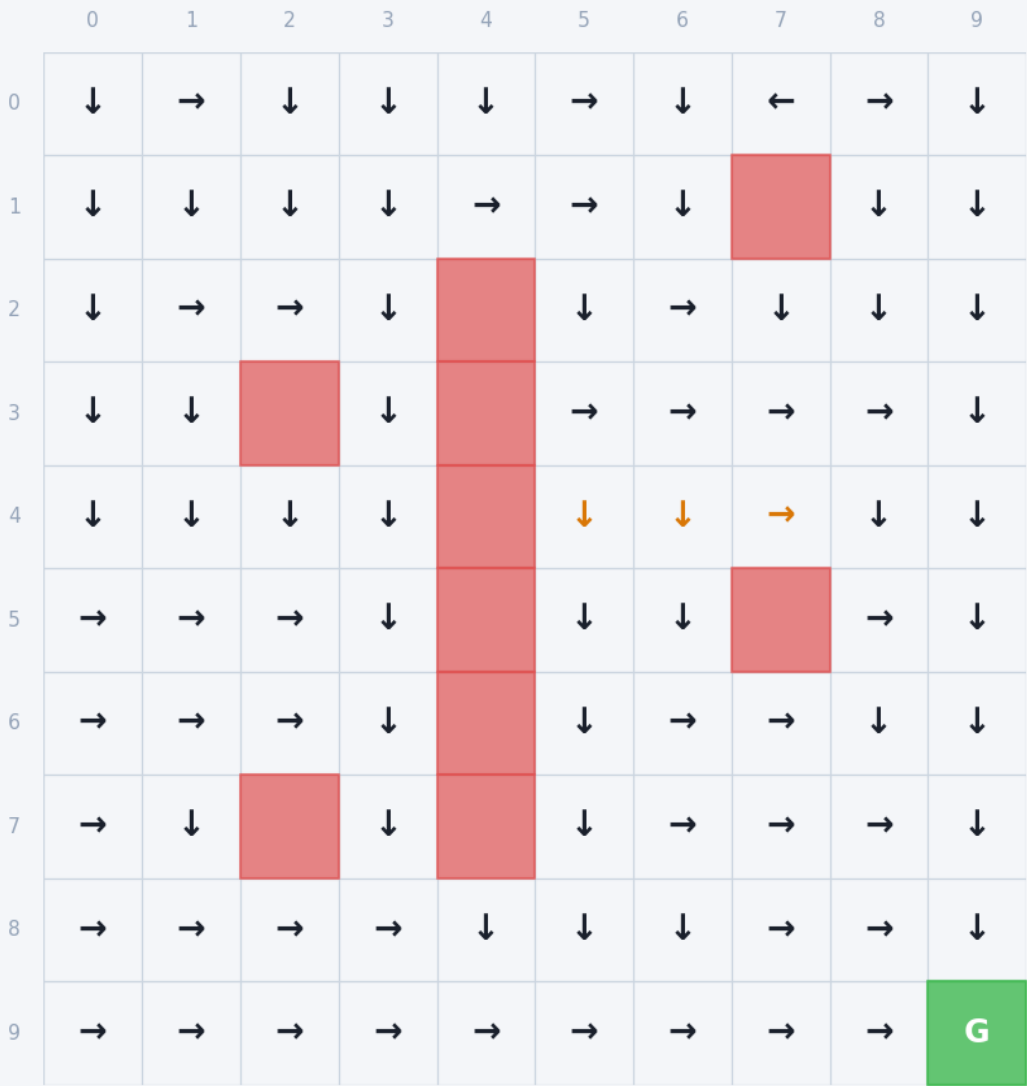
ϵ : 0.05

+86.4

The transformation from blind wandering to purposeful navigation is the numerical signature of reinforcement learning.

Episode 5,000 — What the Agent Learned

100 states. 4 actions each. One number earned the hard way.



Policy distribution

↓ down

47 cells

→ right

41 cells

↑ up

11 cells

← left

1 cell

The dock is at row 9 — go south

The dock is at col 9 — go east

Force detour above obstacle wall

One escape cell — bottom of obstacle

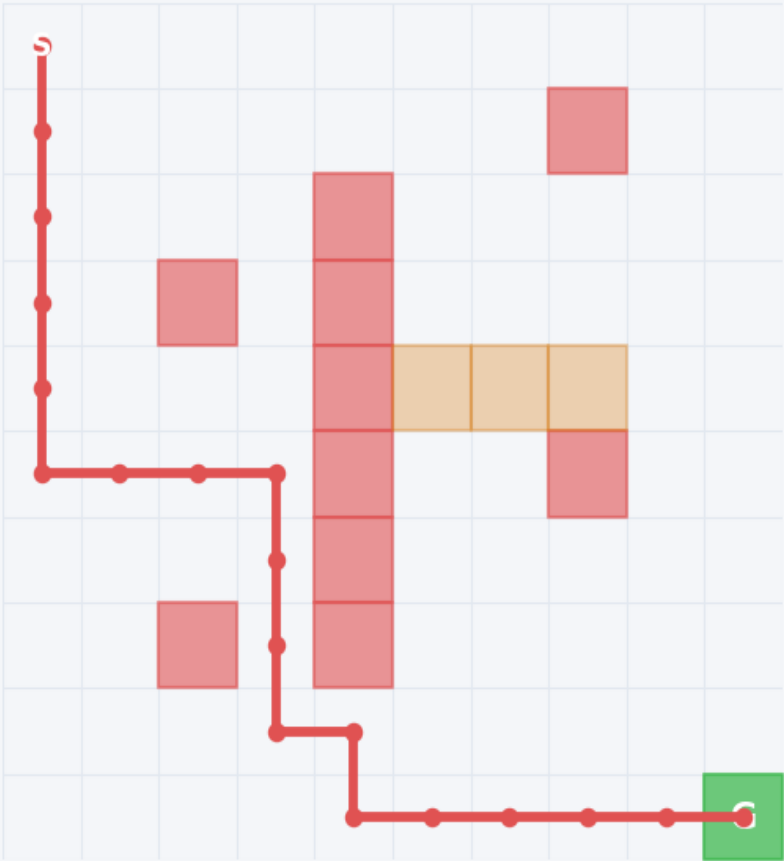
The arrows form a flow field converging on (9,9) — routing around the vertical wall, avoiding congestion where possible.

Three Routes — The Agent in Action

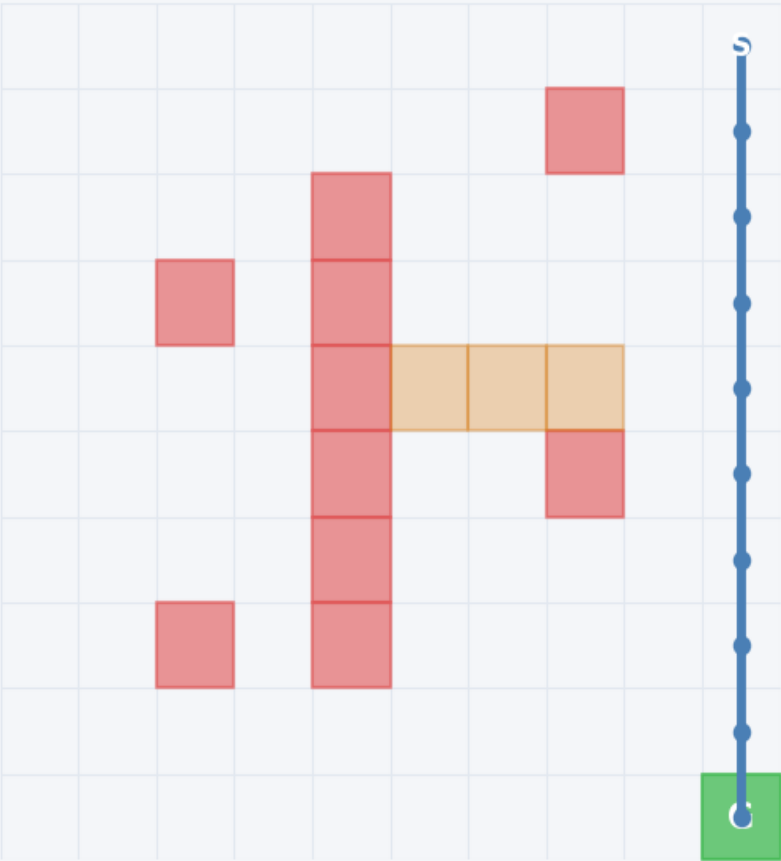
Greedy policy tested from three start positions: always take the action with the highest Q-value.

ROUTES

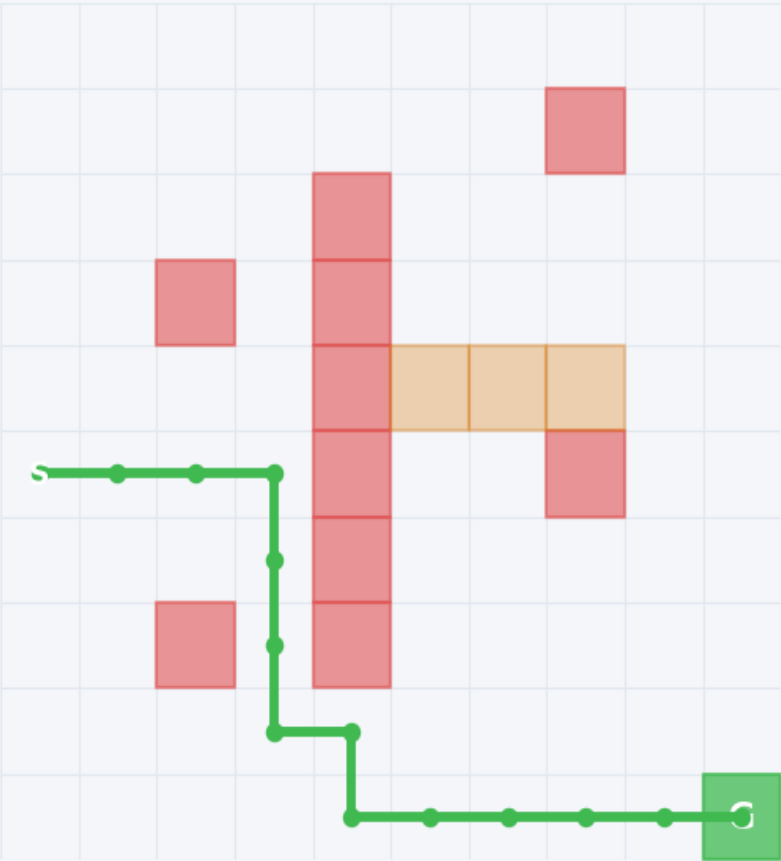
A — Corner-to-Corner
18 steps · Reward: 83



B — Column Sprint
9 steps · Reward: 92



C — Congestion Run
13 steps · Reward: 88



Scenario A · Start (0, 0)

18 steps · Reward 83.0

Column left → cut east below wall → sweep row 9

Scenario B · Start (0, 9)

9 steps · Reward 92.0

Straight down column 9 — no obstacles, no detour

Scenario C · Start (5, 0)

13 steps · Reward 88.0

Skirts congestion zone — trades 1 extra step for -10 avoided

Simulator & Hyperparameter Sensitivity

evaluate_start(row, col) → optimal route, step count, total expected reward.

A — Corner to Corner

Start: (0, 0)

18 steps

Reward: 83.0

19-cell path down left column, cut east, sweep bottom row

B — Column Sprint

Start: (0, 9)

9 steps

Reward: 92.0

Shortest discovered route — straight down clear column 9

C — Congestion Run

Start: (5, 0)

13 steps

Reward: 88.0

Skirts congestion zones — saves -10 penalty per avoided cell

Hyperparameter Sensitivity (last-200 mean reward):

α (learning rate)	α=0.01 83.7	α=0.05 85.2	α=0.10 85.6	α=0.20 86.4	α=0.50 89.6
γ (discount factor)	γ=0.70 86.2	γ=0.80 84.8	γ=0.90 84.0	γ=0.95 85.6	γ=0.99 86.8

γ = 0.95 means a reward 18 steps away (Scenario A) is still worth $95^{18} \approx 40\%$ of face value — critical for planning in a 10×10 grid.

The agent didn't read a manual. It learned by living in the world.

5,000 episodes of consequence. Every wall collision carved knowledge. Every goal arrival confirmed it. By episode 5,000, the agent navigates 18 cells with confidence — not because it was told the route, but because it earned it.

This is the close of a chapter in this portfolio. Twenty-two projects later, we've crossed the full arc: supervised recognition → unsupervised discovery → reinforcement decision-making. Each paradigm answered a different question about machine intelligence.

Train the agent. Vary the episodes. Change the reward function.

Add a new obstacle. Watch how the policy adapts.

Because intelligence, biological or algorithmic, is learned through consequence.

